Homework 5

1. Problems and applications

3.

a) The price rises by **20%** (midpoint method). Quantity demanded falls by **4%** in the short run (**0.2 x 20%**) and by **14%** in the long run (**0.7 x 20%**).

b) Over time, people can insulate their homes or switch to other fuels, so demand becomes more elastic.

4.

inelastic. The price rose and total revenue increased, so the price rose by a larger percentage than quantity fell.

9.

Walt: perfectly inelastic (elasticity = **0**). Jessie: unit elastic (elasticity = **1**), because she always spends **$10**.

10.

a) the price must rise by **20% / 0.4 = 50%**, from **$5** to about **$8.33** (midpoint method).

b) five years from now, because demand is more elastic in the long run.

c) Teenagers have less income and are less addicted, so they react more to price.

11.

if demand is inelastic, increase the price; if elastic, decrease it. Museum demand is usually inelastic (few substitutes), so raising the price will likely increase revenue.

2. Handbook

1-a, 2-b, 3-d, 4-d, 5-d, 6-b, 7-d, 8-b, 9-a, 10-c, 11-c, 12-c, 13-a, 14-a, 15-c, 16-c, 17-a, 18-d, 19-a, 20-a

3. Short essay

A new, more productive wheat hybrid shifts the supply curve to the right, so the price of wheat falls and quantity rises. Since demand for wheat is inelastic, the price falls by more than the quantity rises, and farmers' total revenue decreases. Each farmer still adopts the hybrid, but together they end up earning less. This is why good news for farming can be bad news for farmers.

